# NexStore SaaS Platform

MULTI-TENANT E-COMMERCE ENGINEERING & ARCHITECTURE REPORT

RELEASE STATUS: PRODUCTION READY

Date: **October 25, 2026**
Branch: **main (commit 5c586f0)**
Target: **Enterprise Multi-Tenant SaaS**

| ACTIVE TENANTS | MONTHLY RUN RATE (MRR) | ANNUAL RUN RATE (ARR) | GOVERNMENT COMPLIANCE |
|---|---|---|---|
| **28 Stores** | **BDT 98,000** | **BDT 1.17M** | **DBID & NBR** |
| 100% Operational Isolation | Growth Pro Tier Primary | 100% Retained Revenue | Verified Tax Invoice Engine |

## 01 Dedicated Merchant Profiles & Deep-Dive Command Modal

A multi-tabbed command interface was implemented inside the Super-Admin Command Portal allowing granular, store-by-store management across all 28 registered tenant merchants.

| MANAGEMENT TAB | FEATURES & ARCHITECTURAL CAPABILITIES | PRIMARY CONTROLS & FIELDS |
|---|---|---|
| **1. Subscription & Billing** | Real-time active days counter, renewal countdown, plan tier switching, payment ledger. | Starter (1.5k), Growth Pro (3.5k), Enterprise (8.5k), Add Invoice Form, Status toggle. |
| **2. Feature Gates & Quotas** | Architectural switches and capacity constraints per tenant instance. | Max Products (500), Max Orders (1500), Staff limit (5), CNAME Custom Domain, Courier API. |
| **3. KYC & Legal Verification** | Statutory e-commerce compliance conforming to Bangladesh Ministry of Commerce guidelines. | DBID Registration No, Municipal Trade License, Owner NID/Passport, Commercial Address. |
| **4. Demands & Notes** | Custom feature demand queue & private internal administrator CRM notes. | Feature request logger, Mark fulfilled/delete, Private CRM notes with encrypted storage. |

## 02 Merchant Self-Service "Subscription & Limits" Portal

Merchants can now monitor their platform quotas and billing transparency directly from the merchant dashboard (`/admin/dashboard?tenant=smartvolt`).

### Live Usage & Quota Telemetry Meters

REAL-TIME DB SYNC

- **Product Catalog Meter:** Displays active products vs allowed quota (e.g., 9 / 500 items used on Growth Pro).
- **Monthly Order Meter:** Tracks volume against plan allowances (e.g., 2 / 1,500 monthly transactions).
- **Staff Access Meter:** Monitors provisioned team logins against plan caps (e.g., 1 / 5 staff members).
- **Database Partition:** Visualizes isolated PostgreSQL JSON schema partition health for high throughput.
- **Feature Entitlements Checklist:** Visual green badges for active entitlements and lock badges for restricted features.
- **Upgrade Request Channel:** Merchants can directly submit custom feature demands to Super-Admin.

## 03 Backend Architectural Security Gates & Quota Enforcement

Server-side validation checks were introduced to prevent unauthorized API utilization, resource exhaustion, and suspended account activity.

| MODULE & ROUTE | ENFORCEMENT CONDITION | ERROR / RESTRICTION ACTION |
|---|---|---|
| `catalog.routes.ts` | Subscription status is *suspended* or current product count ≥ `features.maxProducts` | Returns `403 Forbidden` preventing new catalog additions. |
| `order.routes.ts` | Consignment dispatch called when `features.courierLogisticsAPI === false` | Blocks Steadfast / Pathao API dispatch and prompts merchant to upgrade plan. |
| `tenant.routes.ts` | Custom CNAME domain assigned when `features.customDomain === false` | Restricts branded domain pointing and preserves fallback subdomain. |

## 04 Super-Admin Financial & MRR Telemetry Engine

The Super-Admin Central Command Center was enriched with dynamic recurring revenue analytics and retention tracking.

| MONTHLY MRR | ANNUAL ARR | PAYING STORES | UPCOMING DUE |
|---|---|---|---|
| **BDT 98,000** | **BDT 1,176,000** | **28 Stores** | **30-Day Alert** |
| Dynamic Sum | 12x Multiplier | Growth Pro Tier | Direct Manage |

- **Automated MRR Aggregation:** Calculates real-time recurring sales across all active tenant partitions.
- **Tier Penetration Visualizer:** Tracks customer concentration across Starter, Growth Pro, and Enterprise packages.
- **Renewal Pipeline:** Renders an actionable queue of stores sorted by renewal proximity (e.g., SmartVolt renewing in 21 days) with 1-click management shortcuts.

## 05 Official Printable SaaS Invoices & NBR Tax Receipts

A reusable, print-ready corporate invoice modal ( `SaaSInvoiceModal.tsx` ) was constructed for regulatory compliance and audit documentation.

### Statutory Receipt Specifications

NBR COMPLIANT

- **Platform Provider Header:** Silicon Tower, Gulshan-2, Dhaka with Central Helpline `09606 111 777` and official VAT/BIN ID `9928172948-0201` .
- **Merchant Verification:** Billed-to card featuring Store Name, Subdomain, Principal Owner, and Verified DBID registration number.
- **Itemized Financial Breakdown:** Cloud hosting, PostgreSQL partitioned database, anti-fraud engine, and SLA guarantees itemized at BDT 3,500.
- **Verification Seal:** Green electronic "PAYMENT RECEIVED & VERIFIED" badge with clearing method (bKash Merchant / Bank Wire).
- **Print Engine:** Embedded `@media print` rules supporting instant A4 paper printing and vector PDF export.

## 06 Storefront Consumer Protection & SEO Infrastructure

- **Consumer Rights Policies:** 3 distinct statutory pages ( `/privacy-policy` , `/refund-policy` , `/terms-and-conditions` ) ensuring full compliance with national e-commerce guidelines.
- **SEO Infrastructure:** Dynamic `sitemap.ts` and `robots.ts` with priority crawl instructions for rapid search indexing.
- **Customer Engagement:** Floating WhatsApp assistance widget with contextual messages and centralized hotline support.
- **Localization Cleanup:** English standardization completed across all storefront footers, checkout manifests, and admin forms.